Kartik Gonde

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Education

University of Texas, Dallas, TX **Aug 2024 – Present**
Master of Science in Computer Science GPA: 3.50/4.0
Relevant Coursework: Design and Analysis of Algorithms, Database Design, Big Data Management and Analytics, Natural Language Processing

Savitrbai Phule Pune University, India **July 2020 – May 2024**
Bachelor of Engineering in Computer Engineering Cumulative GPA: 9.1/10.0
Relevant Coursework: Data Structures and Algorithms, Machine Learning, Operating Systems, Distributed Computing, Cloud Computing

Professional Experience

Data Engineer Intern *Zensar Technologies* **Feb 2024 - May 2024**

- Engineered and deployed 3 interactive Power BI dashboards by integrating real-time data from SQL and cloud sources, enabling cross-functional stakeholders to monitor KPIs dynamically and driving a 10% uplift in customer satisfaction.
- Streamlined backend operations by developing Python and SQL scripts, leading to a 13% improvement in data processing efficiency.
- Implemented and managed CI/CD pipelines with Jenkins to streamline deployment processes and minimize manual errors.

Machine Learning Research Intern *Indian Institute of Technology Ropar, Punjab, India* **Jan 2023 - Aug 2023**

- Designed a state-of-the-art equivalent deep learning model for the task of "image restoration."
- The research conducted during the internship was accepted for presentation at the IEEE Conference on Advanced Video and Signal Based Surveillance (AVSS) 2024, held in Niagara Falls, Canada.
- Collaborated with PhD researchers and faculty to design and fine-tune experimental pipelines, contributing to publications and real-world deployment readiness.

Software Engineer Intern *Propellum Infotech* **Aug 2022 - Nov 2022**

- Redesigned an enterprise-grade Angular app, translating Figma designs into functional components and improving user flows by 50%.
- Developed and integrated 10+ Angular components, landing pages, and React micro-frontends with Webpack5 module federation.
- Streamlined backend operations by developing Python and SQL scripts, leading to a 13% improvement in data processing efficiency.

Publications

AeroDehazeNet: Exploiting Selective Multi-Scale Transformers for Aerial Image Dehazing **18 Sept 2024**
(Published)

- Developed a transformer-based network (AeroDehazeNet) for aerial image dehazing in remote sensing applications using PyTorch and Tensorflow.
- Proposed a multi-scale attention network coupled with a context enhancement block, utilized as a feed-forward mechanism, to effectively capture intricate image details.
- Reduced the parameter count from 20.32 million to 16.61 million.
- Achieved strong SSIM and PSNR results on both synthetic and real-world datasets while utilizing fewer FLOPs.

Skills

Languages	:	HTML/CSS3, Javascript(ES6), Python, C++
Web Technologies	:	React.js, Node.js, Angular, Lit, Shoelace, Sass/SCSS, Postman
Databases	:	MySQL, MongoDB, SQL
Big Data Technologies	:	Apache Spark, Hive, Databricks
Collaboration Tools	:	GIT, Jira, ServiceNow
Machine Learning	:	PyTorch, Tensorflow, Scikit-Learn, NLTK, OpenCV, Pandas, NumPy

Projects

Spark Streaming with Real Time Data and Kafka **Mar 2025**

- Developed an end-to-end real-time data pipeline that streams Reddit comments using PRAW and ingests them into Apache Kafka.
- Designed a PySpark Structured Streaming consumer to extract named entities from text and compute their real-time frequency counts.
- Integrated Elasticsearch and Logstash to index real-time entity data and visualized the top 10 named entities in Kibana using dynamic bar charts.

ChessMate **May 2025**

- Developed a real-time two-player chess game using Node.js and Express.js, implementing WebSocket communication through Socket.IO for instant game state synchronization between players
- Built a responsive frontend interface using EJS templating engine and vanilla JavaScript, creating an intuitive user experience for players and spectators

Chat Flow **June 2025**

- Developed a full-stack real-time chat application using the MERN stack (MongoDB, Express.js, React, Node.js) with Vite for fast development and optimized builds.
- Built secure authentication and authorization flows using JSON Web Tokens (JWT) and developed RESTful APIs for user management, authentication, and messaging.